# *8:45-90:00. The Influence of a High Fat Meal Compared to an Olestra Meal on Coronary Artery Endothelial Dysfunction by Rubidium (Rb)-82 Positron Em...*

Background: Cellular changes lead to coronary artery endothelial dysfunction (ED) and precede plaque formation. Clinical events, such as unstable angina and acute coronary syndromes, are common consequences of ED. Coronary artery ED, as characterized by Rb-82 PE, is a perfusion abnormality at rest, which improves following stress. In risk factor modification studies, particularly in cholesterol-lowering trials, coronary artery ED has been demonstrated to be reversible. Other studies have correlated low fat diet modification with improvement in coronary artery disease.Purpose: This study evaluates changes in myocardial perfusion following meals with low versus high TG content, and its influence on post prandial serum TG.Methods: With a randomized, double blind placebo controlled, cross over design, we investigated 19 patients (10 with ED and 9 with normal perfusion) with Rb-82 PET for myocardial blood flow at rest and with adenosine stress. PET images and serum triglycerides were obtained before and after an olestra (OA) meal (2.7g TG, 44g olestra) and a high-fat meal (46.7g TG). Meals were matched for carbohydrate, protein, and cholesterol content.Results: Myocardial perfusion (uCi/cc) increased 11 - 12% following the OA meal compared to the high-fat meal in patients with ED. For all patients combined, serum TG increased significantly ($p < 0.01$) in the non-OA group with the median change from baseline to 170.0 mg/dl, compared to 21.5 mg/dl in the OA group during the 6 hours following the meal.Conclusions: A single olestra meal significantly diminishes post prandial serum TG levels and improves myocardial perfusion in patients with endothelial disease.